

Verifying Age Online Without Sacrificing Privacy

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As governments and online platforms seek to better protect minors from harmful or age-restricted content, age verification has become an increasingly important topic. While the objective appears straightforward, determining a user's age online presents significant challenges. Any solution must balance several competing requirements: privacy, security, usability, and reliability.

This project examined the technologies currently used for online age verification and evaluated the risks and benefits associated with each approach. The research focused particularly on privacy-preserving methods and the emerging European framework for digital identity and age verification. The findings suggest that although age verification can improve online safety, no single solution fully satisfies all requirements without introducing trade-offs.

Current Approaches to Age Verification

Several methods are currently used or proposed to verify a user's age online.

The most basic approach is self-declaration, where users simply enter their date of birth. While easy to implement and convenient for users, this method offers little assurance because it can easily be bypassed.

A more robust approach requires users to submit official identity documents, such as passports or national identity cards. These systems provide a high degree of confidence regarding a user's age but raise important questions about data collection, storage, and privacy.

Another increasingly common method uses artificial intelligence to estimate a person's age from a selfie or video recording. This reduces the need

to share official documents but introduces concerns regarding accuracy, bias, and the handling of biometric information.

A fourth approach relies on digital identity systems. Rather than repeatedly sharing personal documents, users receive trusted digital credentials that can be used to prove specific attributes, such as being over the age of eighteen. This model is receiving significant attention throughout Europe and formed the primary focus of this research.

The Promise and Challenges of Digital Identity

One of the most significant developments in this area is the European Age Verification App. The objective of this initiative is to enable citizens to securely store and present verified age-related credentials when accessing online services.

Instead of uploading identity documents to every platform individually, users would be able to demonstrate that they meet a required age threshold through a trusted digital credential. This approach has clear privacy advantages because it limits the amount of personal information shared with websites and online services.

However, the system also introduces new security challenges. Sensitive information must be processed, stored, and protected against misuse. The effectiveness of such a system depends not only on its privacy protections but also on the security measures used to prevent fraud, credential theft, and unauthorized access.

Lessons from the Financial Sector

To better understand how high-assurance identity verification can be implemented, this research examined technologies already used within the fi-

nancial sector. Banks and financial institutions have long relied on identity verification processes to comply with anti-money laundering and Know Your Customer requirements.

These systems often combine multiple verification techniques, including document validation, biometric comparison, authenticity checks, and verification against trusted government records. Because these processes have been refined over many years, they provide useful insights into how age-verification systems can improve both reliability and security.

The experiences and technologies developed within the financial sector can therefore serve as valuable reference points for future improvements to European age-verification solutions.

Towards Safer Age Verification

Based on the findings of this research, future age-verification systems should place greater emphasis on privacy-preserving technologies while simultaneously strengthening their security mechanisms.

The current implementation of the European Age Verification App demonstrates considerable potential but also leaves room for improvement. In particular, greater assurance is needed regarding the verification process itself and the protection of sensitive user information. Systems that rely heavily on user-controlled inputs or local age-estimation mechanisms may not provide a sufficient level of trust for high-assurance age verification.

As age-verification requirements continue to expand, ensuring both privacy and security will

remain a critical challenge for governments, regulators, and technology providers.

Recommendation: A Web-Based Verification Service

After evaluating the advantages and limitations of the current approach, this research proposes an alternative architecture based on a government-operated web service.

Rather than requiring users to install a dedicated application, age verification could be performed through a secure government-managed website. Users would authenticate themselves using modern technologies such as passkeys, which provide strong cryptographic authentication while remaining user-friendly.

When visiting a website that requires age verification, users could be redirected to the government verification service. After confirming the required age attribute, the service would provide a verification response without disclosing unnecessary personal information.

A key design objective of this proposal is privacy preservation. The system should be implemented in such a way that the verification authority cannot determine which websites a user visits, thereby preventing the creation of browsing profiles while still enabling reliable age verification.

The internet has traditionally offered users a high degree of privacy, and future age-verification systems should seek to preserve that principle. At the same time, preventing minors from accessing age-restricted content remains a legitimate objective that can be achieved through carefully designed and privacy-conscious verification systems.